Model checking

Wednesday, October 28, 2026

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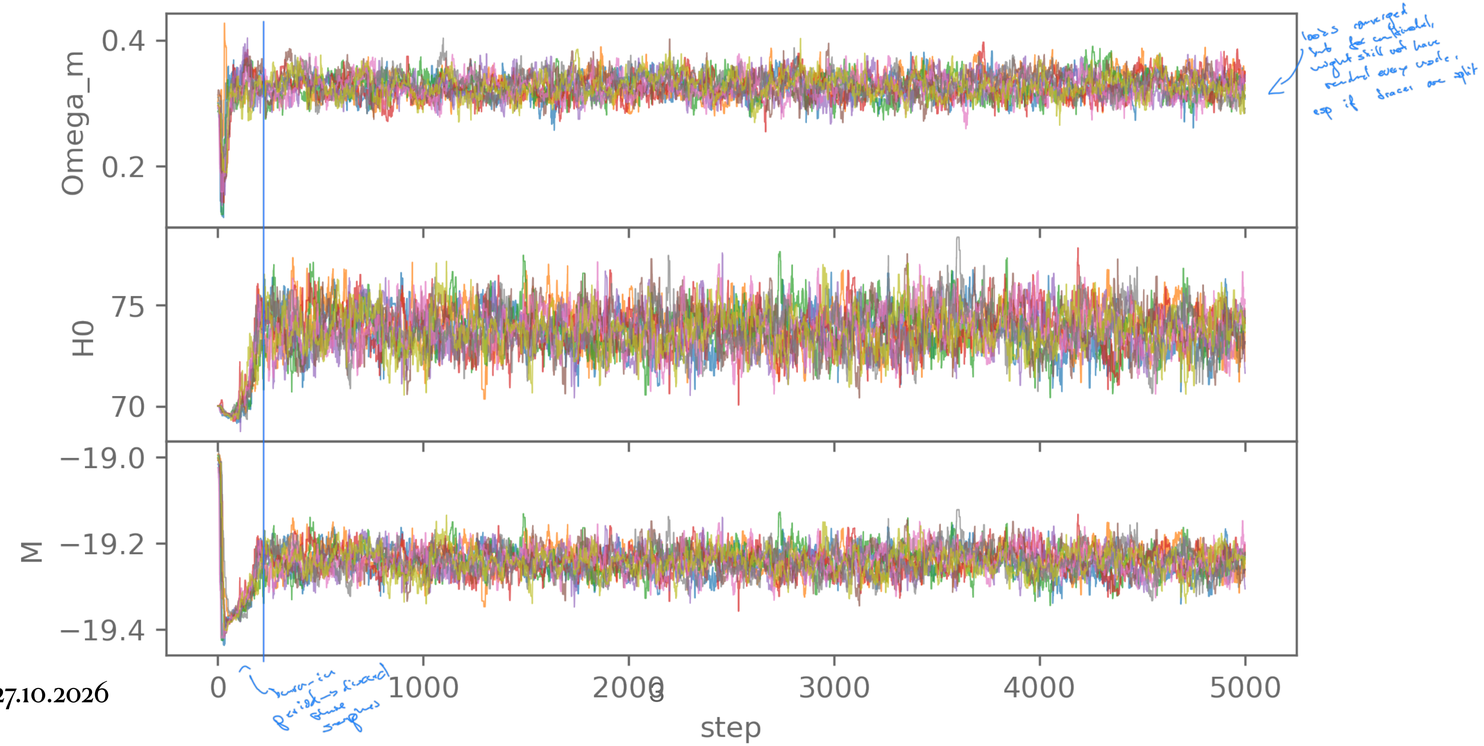
Outline of the day

I've decided to split things up

- How to check chain convergence and autocorrelation
- Interpreting posteriors
- Model checking:
  - Posterior predictive test statistics
- **Next week:** Model selection:
  - Recap: Marginal likelihoods, Bayes factors
  - Leave-one-out (LOO) cross-validation
  - Approximations/alternatives to LOO (WAIC, DIC, etc.)

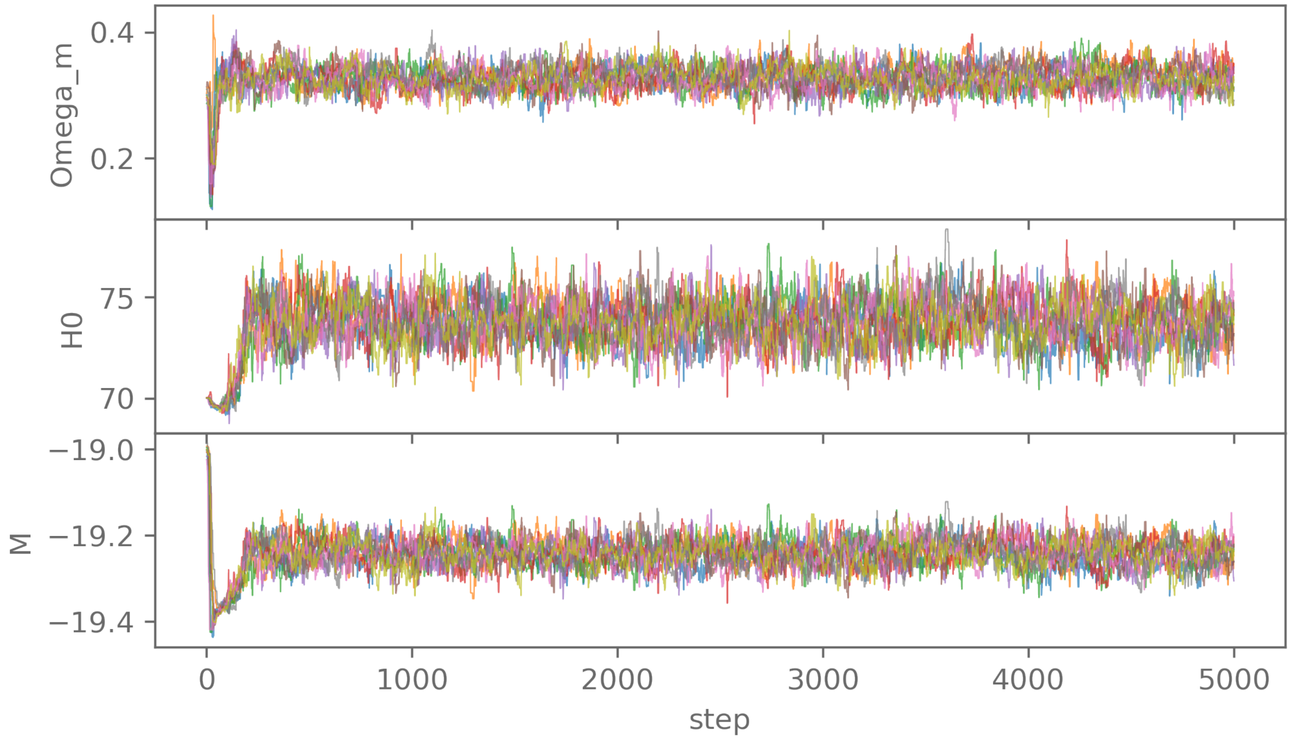
Checking chain convergence

- How do we know when an MCMC chain is converged?
  - **Question:** what are some of the things we might look for?



Checking chain convergence

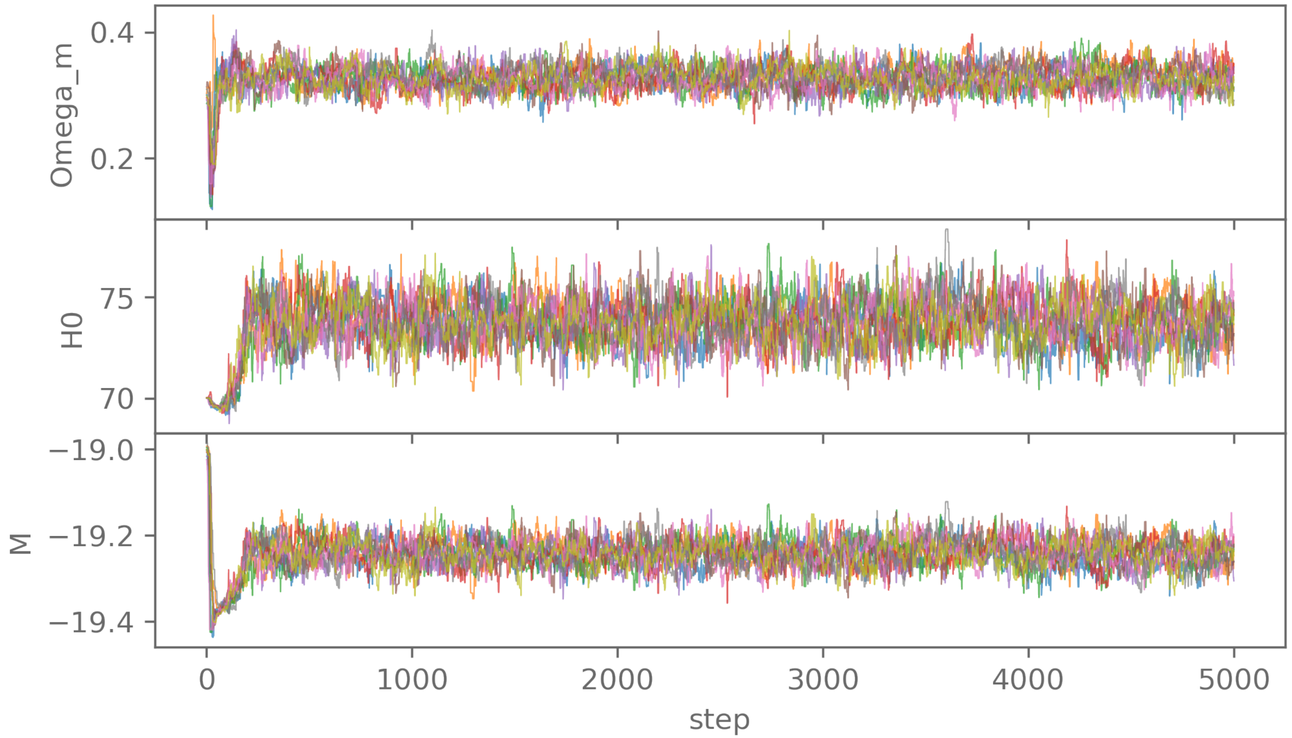
- Features (after burn-in)
  - Mean and variance don't change with time
  - Different "walkers" show similar behavior
- **Gelman-Rubin** statistic formalizes these ideas.



Checking chain convergence

Gelman Rubin Statistic: original paper

- $\hat{R}^2 = \frac{S-1}{S} + \frac{B}{W}$
- Where “S” is the length of our chains.
- B is the **variance** over the **mean** of each chain (average each color, take the variance of that).
- W is the **mean** of the **variances** of each chain (take variance of each color, take mean of that).

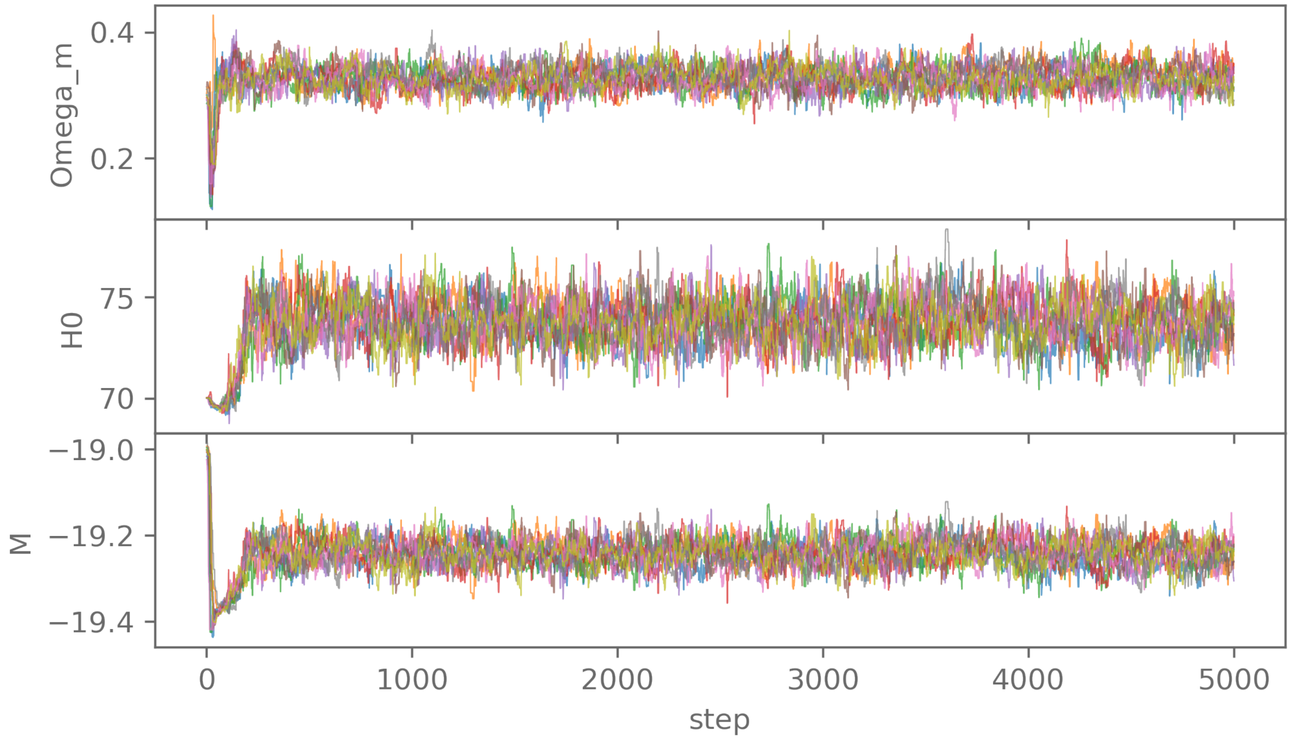


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Checking chain convergence

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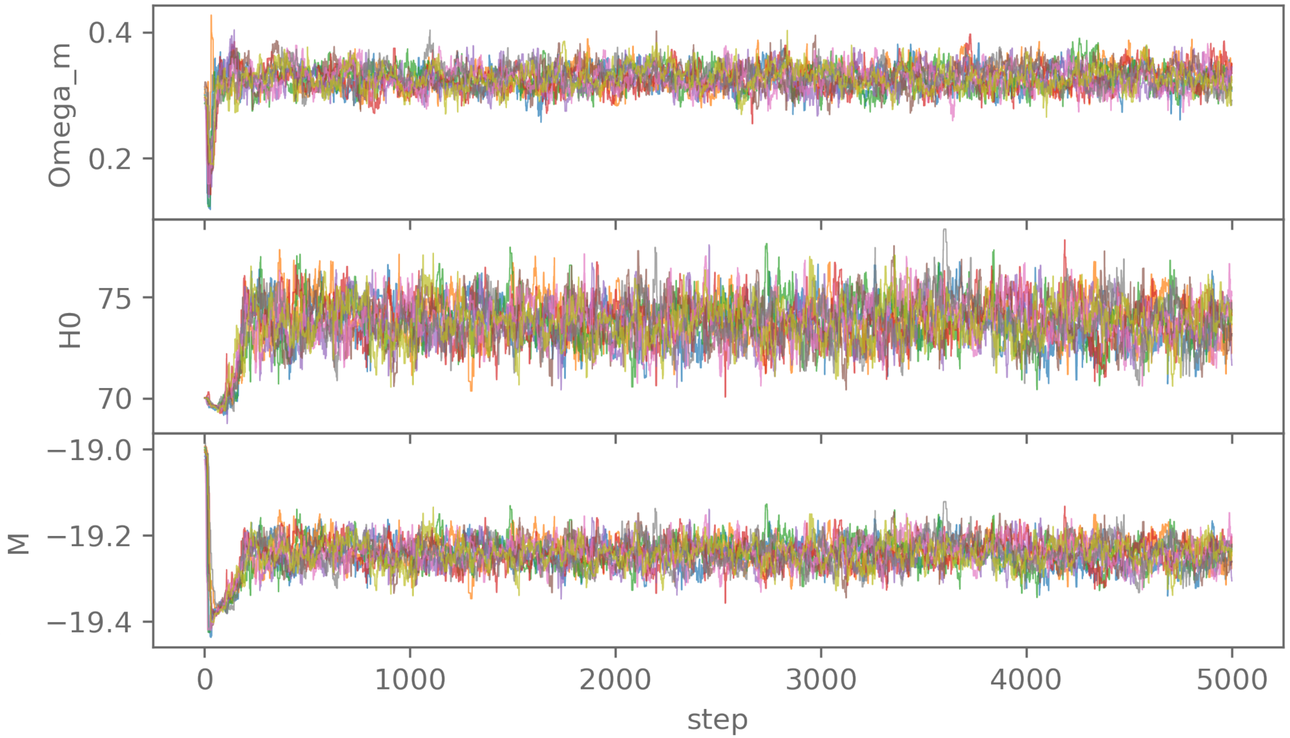
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Checking chain convergence

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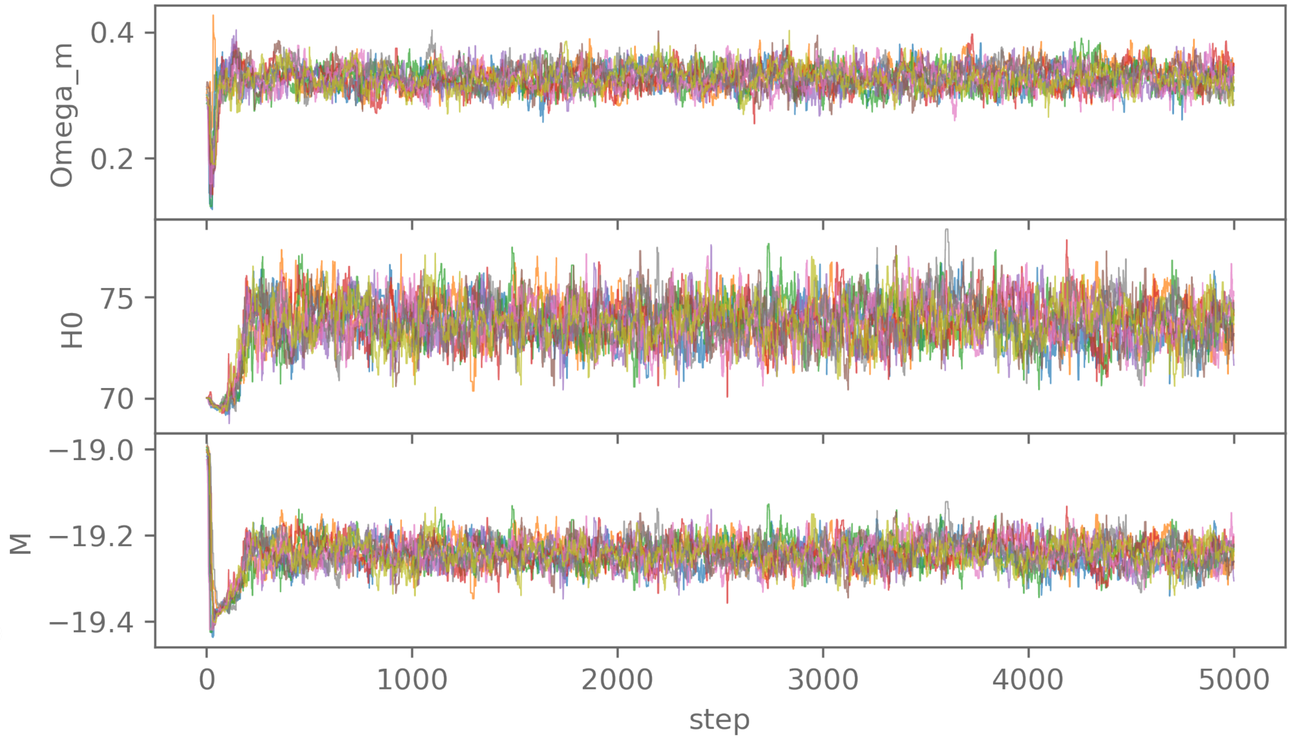
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Checking chain convergence

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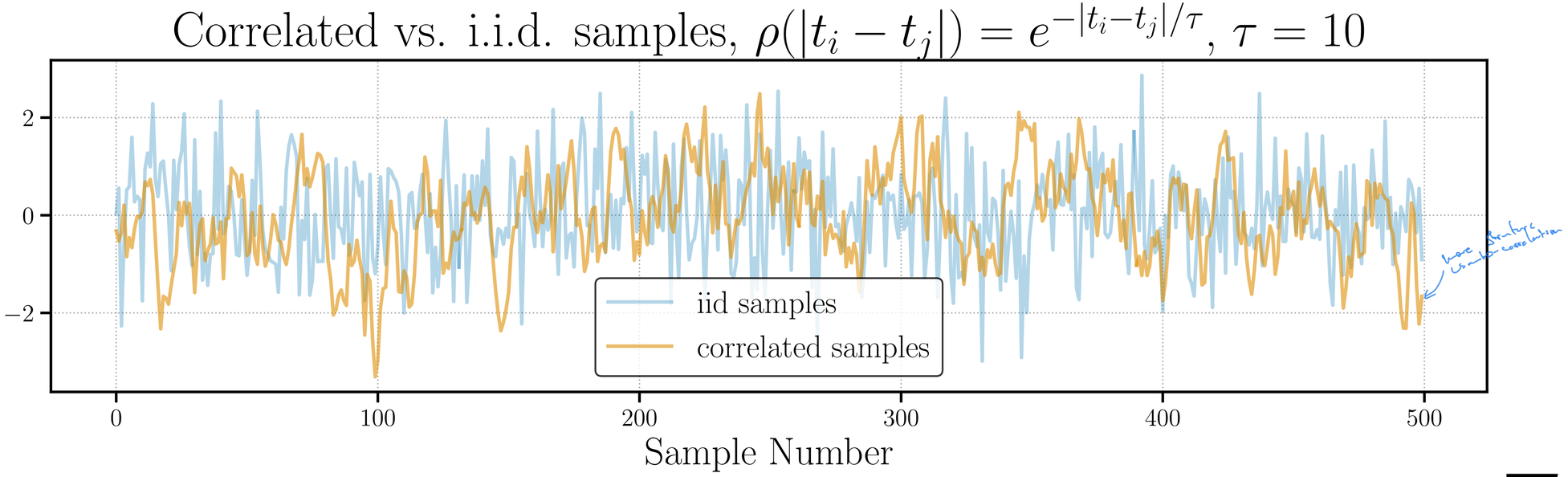
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- Can also be calculated by splitting one long chain into pieces and treating them as separate chains.
  - In general, one asks that $\hat{R} \lesssim 1.01$ or so to claim a chain is converged.



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Autocorrelation and effective sample size

- What happens if our samples are **strongly correlated?**
  - **Consider:** two chains of the same size below. Which one has a better **standard error of the mean:** sqrt(variance divided by length of the chain)?



Autocorrelation and effective sample size

- **Autocorrelation** says how self-similar a chain is—could we use the current state to predict the next state?
- We can calculate a covariance of the chain with a shifted version of itself.

$$\hat{c}_k = \langle (x_i - \bar{x})(x_{i+k} - \bar{x}) \rangle$$

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- **Note:** $\hat{c}_0$ is just simply the variance. The **autocorrelation function** ρ_k is the normalized version of this:

$$\rho_k \equiv c_k / c_0.$$

↳ should Roll-off for MCMC

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  - The **autocorrelation time** is defined as the sum of ρ over all possible (forward and backwards) lags:

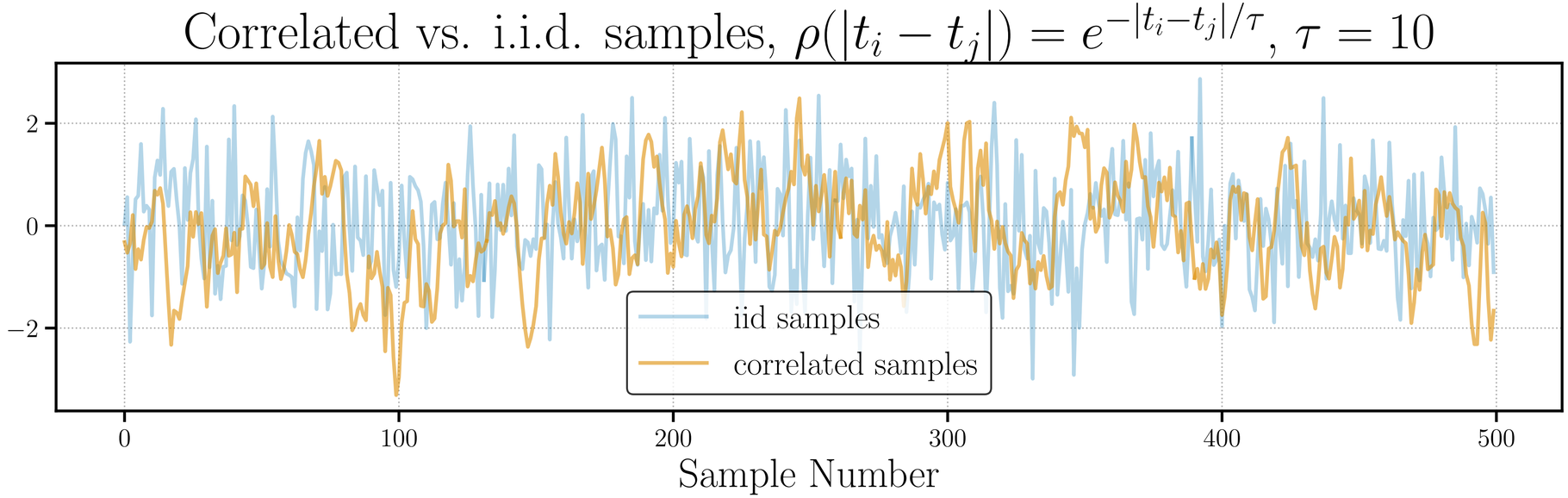
$$\tau = \sum_{k=-\infty}^{k=\infty} \rho_k = 1 + 2 \sum_{k=1}^{\infty} \rho_k$$

symmetric in same density

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Autocorrelation and effective sample size

- **The autocorrelation time** tells us the time-scale on which the chain looks "self-similar."
- If I take two points separated by τ, then I should not be able to predict the second point based on the first.



Autocorrelation and effective sample size

- Additionally, if you want an estimate of the **mean** of your chain, then the variance on that mean depends on the autocorrelation time.

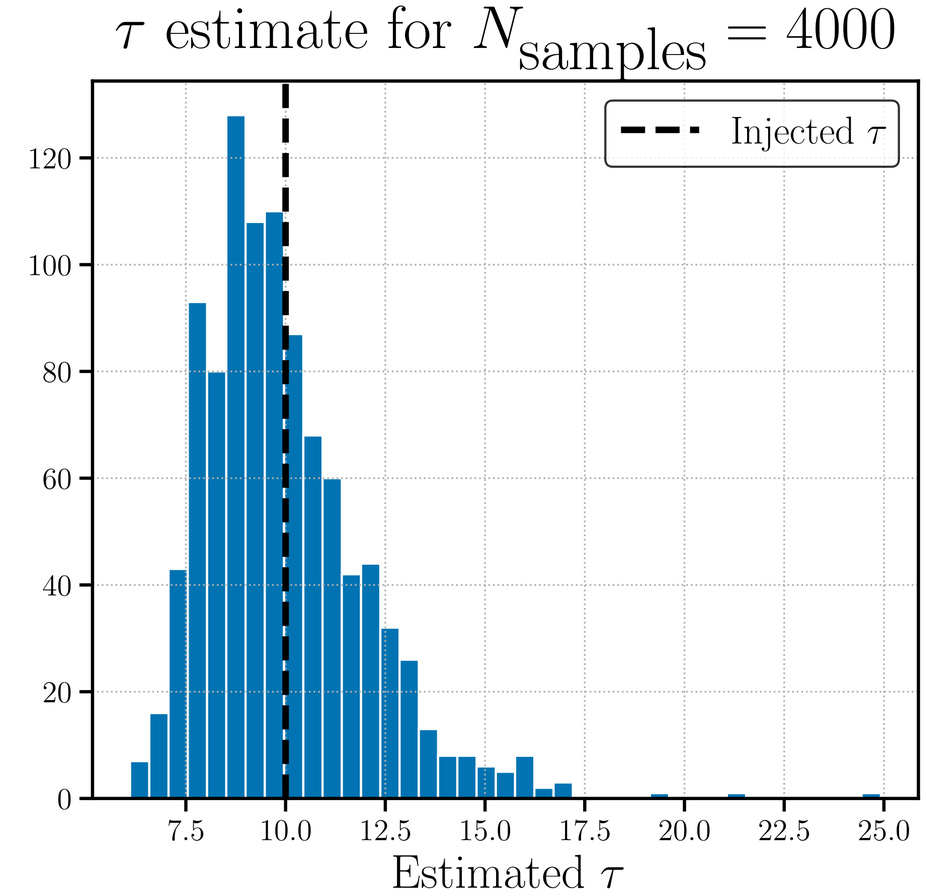
$$\bar{x} = \frac{1}{S}\sum_i x_i \qquad \sigma^2_{\bar{x}} = \frac{\tau}{S}\sigma^2 = \frac{\tau}{S^2}\sum_i (x_i - \bar{x})^2$$

- For **uncorrelated, i.i.d.** samples, $\tau = 1$ and we recover the typical *standard error on the mean*.
- This then suggests a definition of an **effective sample size** $n_{eff} \equiv S/\tau$
- This effective sample size is another useful tool for checking that we explored our parameter space

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Autocorrelation and effective sample size

- This effective sample size is another useful tool for checking that we explored our parameter space, and we have explored our space well.
- Typically we want **at least** $n_{eff} \gtrsim \mathcal{O}(100)$.
- Often people will "thin" a chain by the estimated autocorrelation time
- For $N_{\text{samples}} \lesssim 50\tau$, the emcee estimator is often unreliable



emcee's estimator can be noisy.
See discussion here.

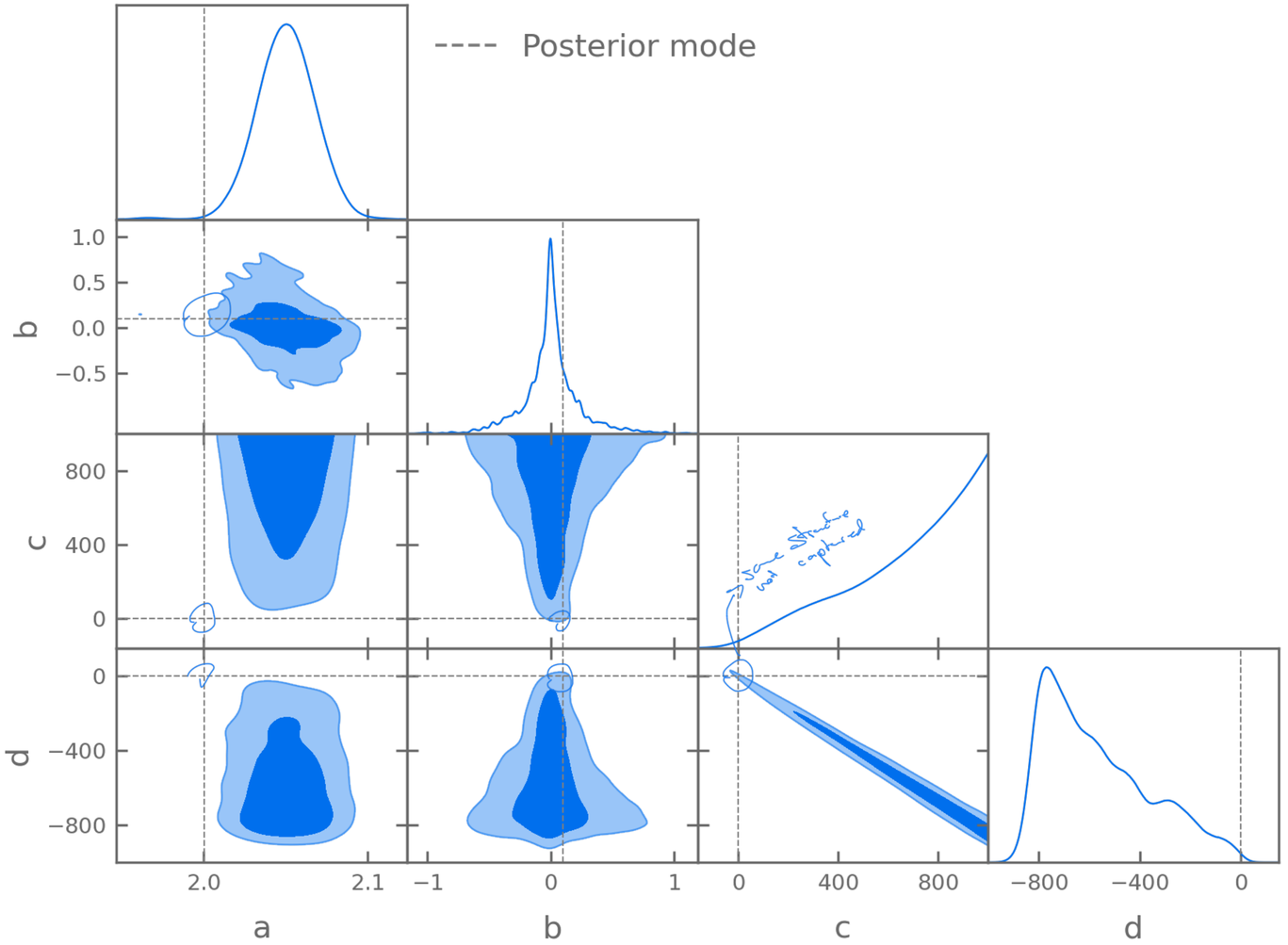
Presenting results

What's the best way?

- When posterior distributions are non-trivial shapes, it can take thought about how best to present results.
- There are a few common ways:
  - **Just show the posteriors!** — e.g., corner plots (only show 1 and 2D marginals)
  - Credible intervals — e.g., 90% credible intervals for 1-dimensional marginals

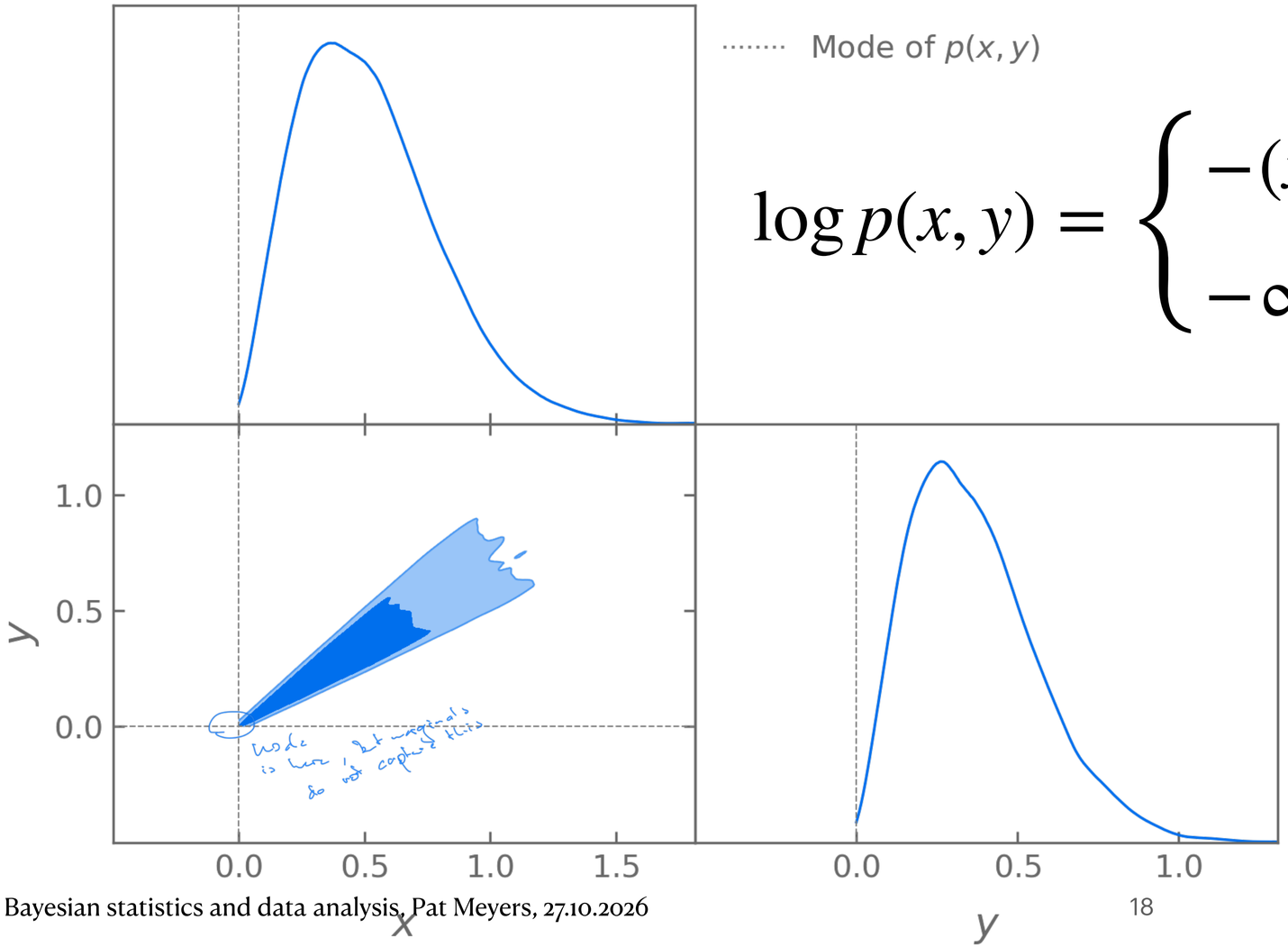
Projection effects — when marginals fail

- The model here is a Gaussian likelihood with mean:
- $\mu = a + bx\left(\frac{1+x}{1+x_0}\right)^c\left(\frac{1+x^2}{1+x_0^2}\right)^d$
- With $x_0 = 0.5$
- When $b \approx 0$, c and d are effectively unconstrained.
- Depending on how the prior is chosen, this can lead to large volumes of posterior space where $b \approx 0$
- Projections will then make our marginals not centered on the actual mode of the distribution!



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Projection effects — when marginals fail



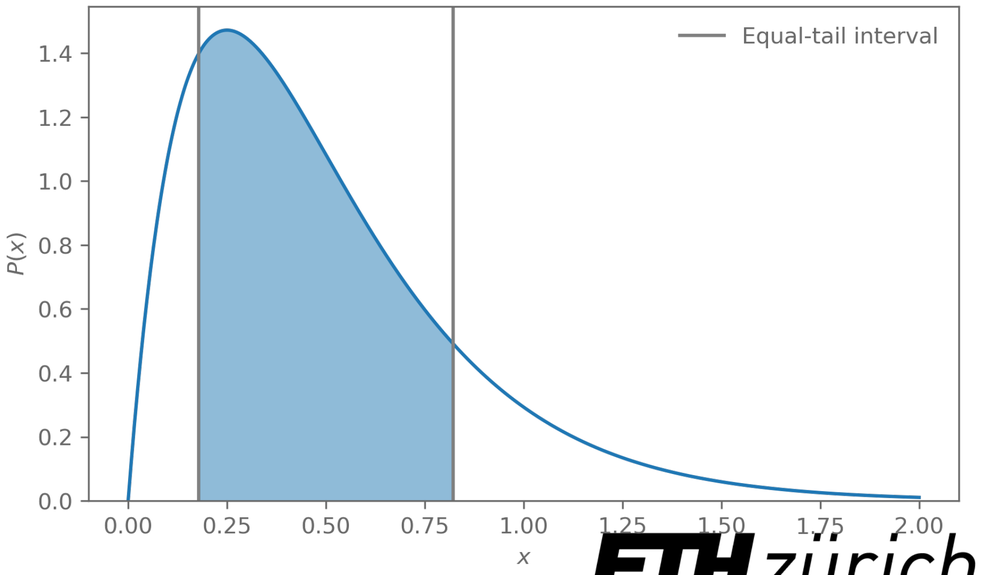
$$\log p(x, y) = \begin{cases} -(x+y)^2 & \text{if } x, y \geq 0 \text{ and } \frac{x}{2} \leq y \leq x \\ -\infty & \text{otherwise.} \end{cases}$$

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Credible intervals — equal tail interval

- In principle there are infinitely many ways to produce a credible interval with, e.g. 90% of the probability under it.
- We commonly use an “equal tail interval” where each tail contains $\alpha/2$ and the interval contains $1 - \alpha$.
- We can calculate this with just the percentile of our samples

```
fifth, fiftieth, ninety_fifth = np.percentile(iid_samples, [5, 50, 95])
print(fifth, fiftieth, ninety_fifth)
# -1.678456534477638 0.011623003859333343 1.6180684508496477
```



Credible intervals — highest density interval

- This is the region (or regions) for which $p(x)$ is above some threshold c_α, such that

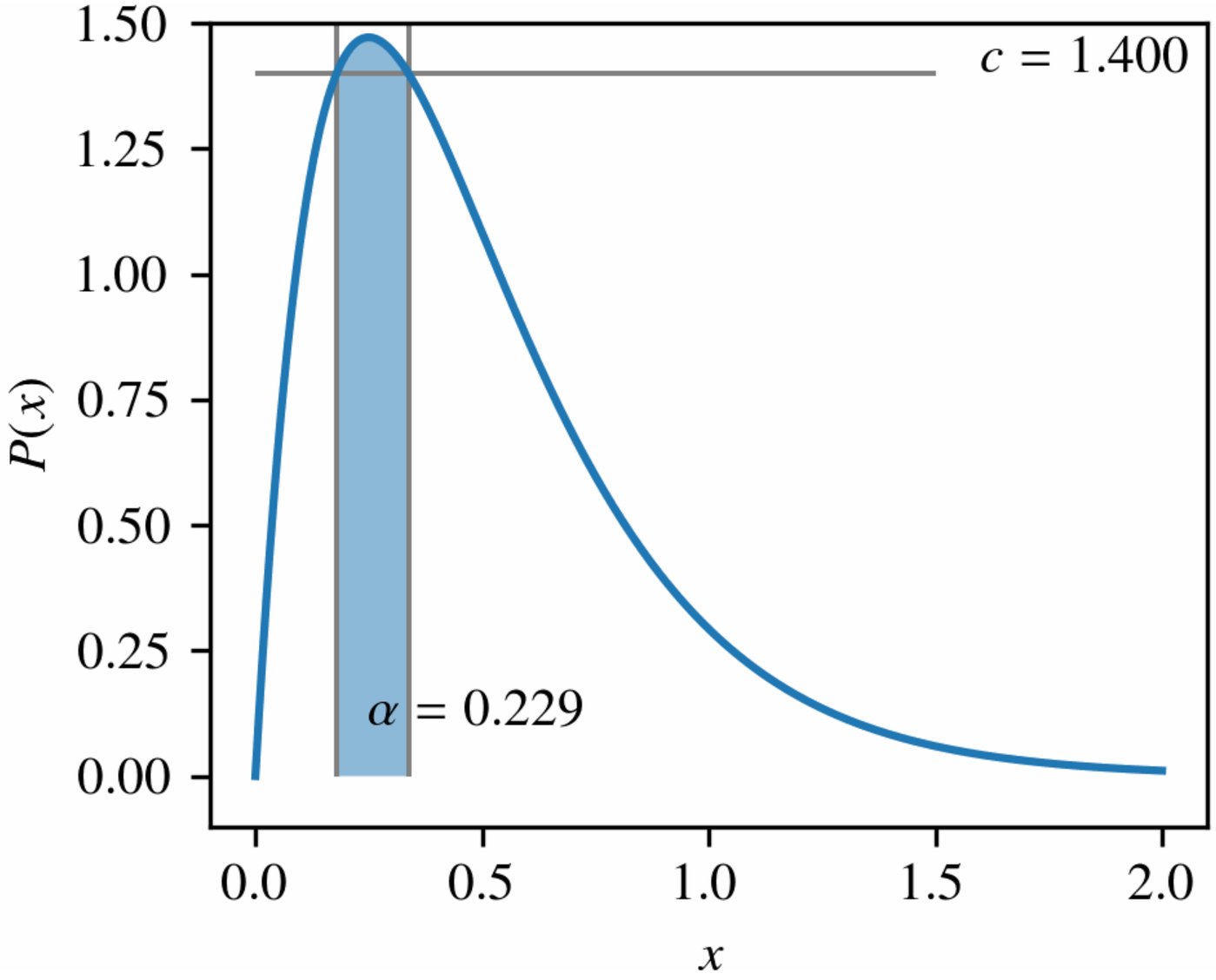
$$\int_{\Omega(\alpha)} p(x)\,dx = 1 - \alpha$$

- where

$$\Omega(\alpha) \equiv \{x \,|\, p(x) > c_\alpha\}\,.$$

- This is the smallest possible credible region. It can also produce disjoint intervals if the posterior is multi-modal (see exercise).
- It's also harder to compute!

Credible intervals — highest density interval

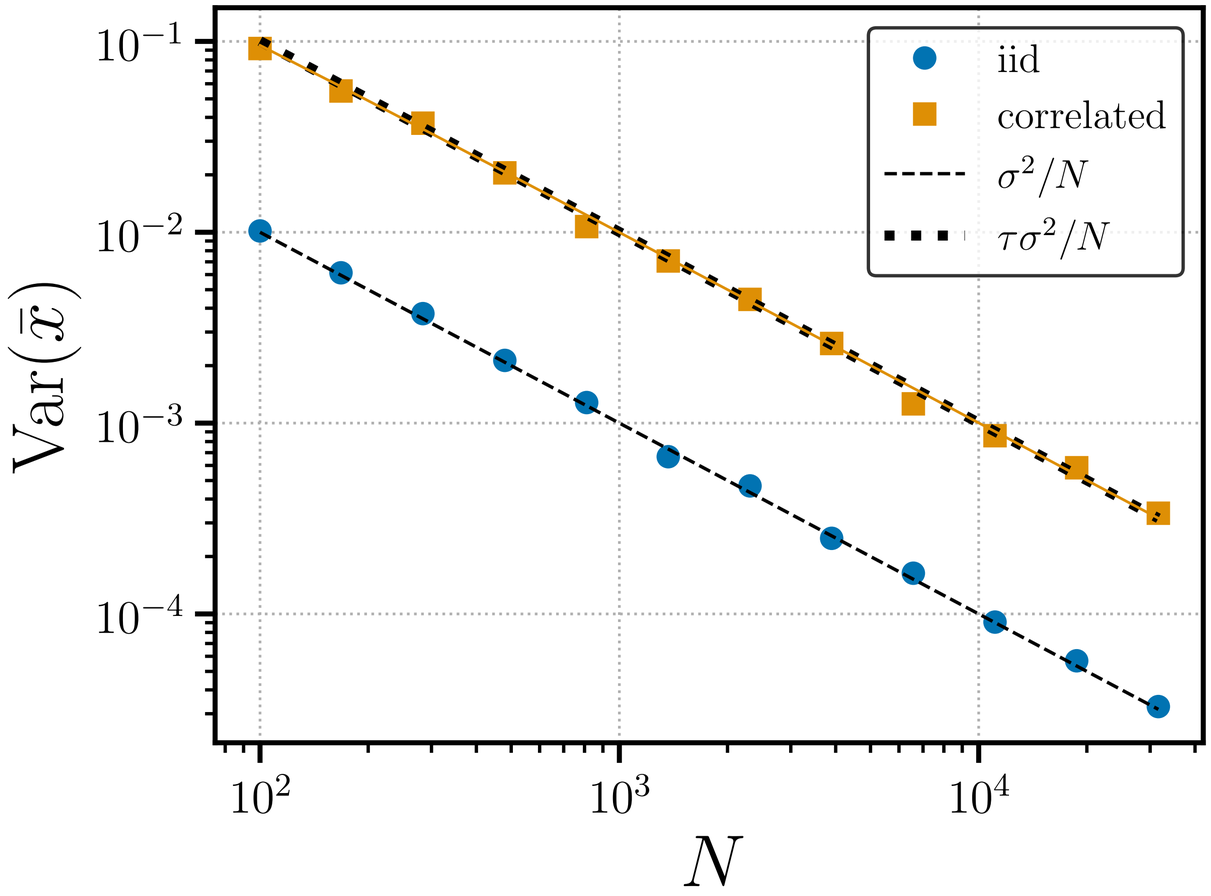


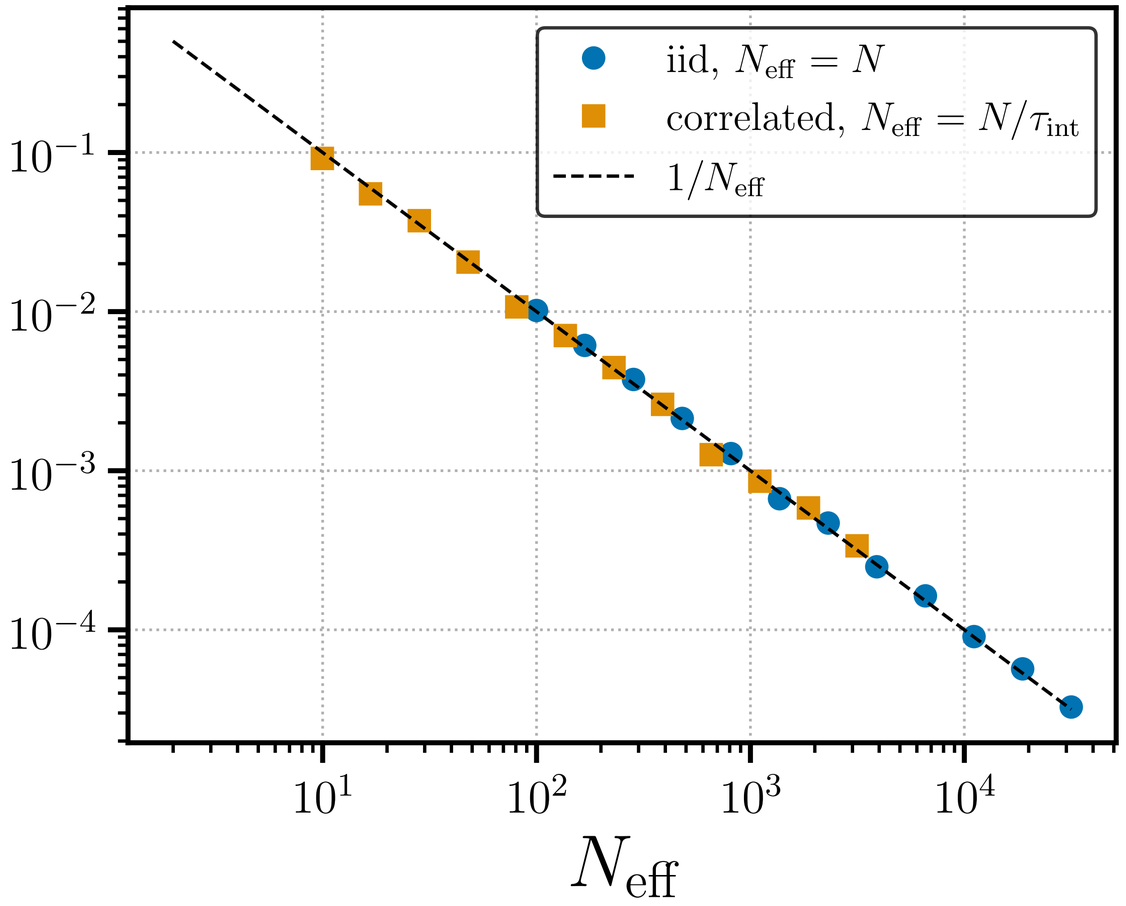
Exercise

- We'll now look at Exercises 1 and 2. See the notebook!

Autocorrelation and effective sample size

Example

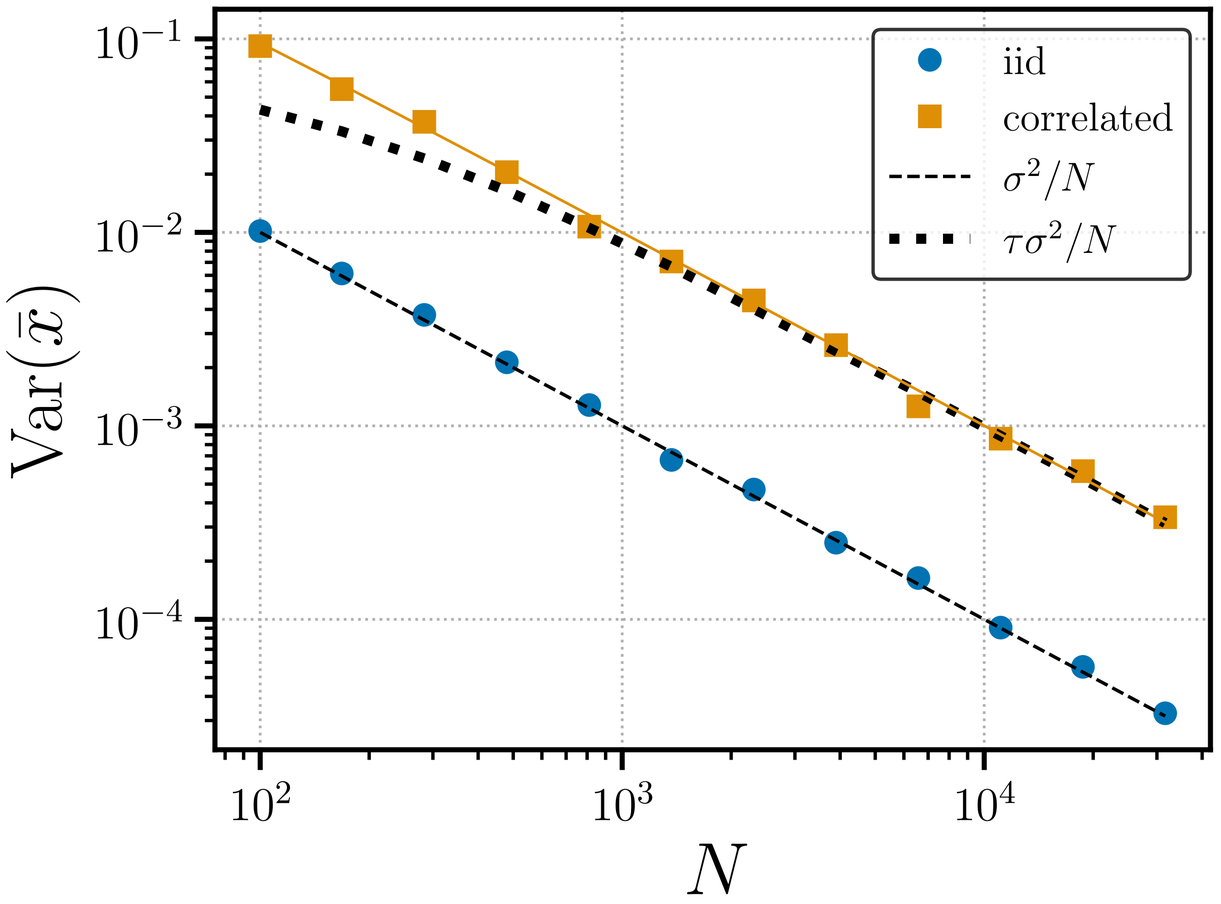


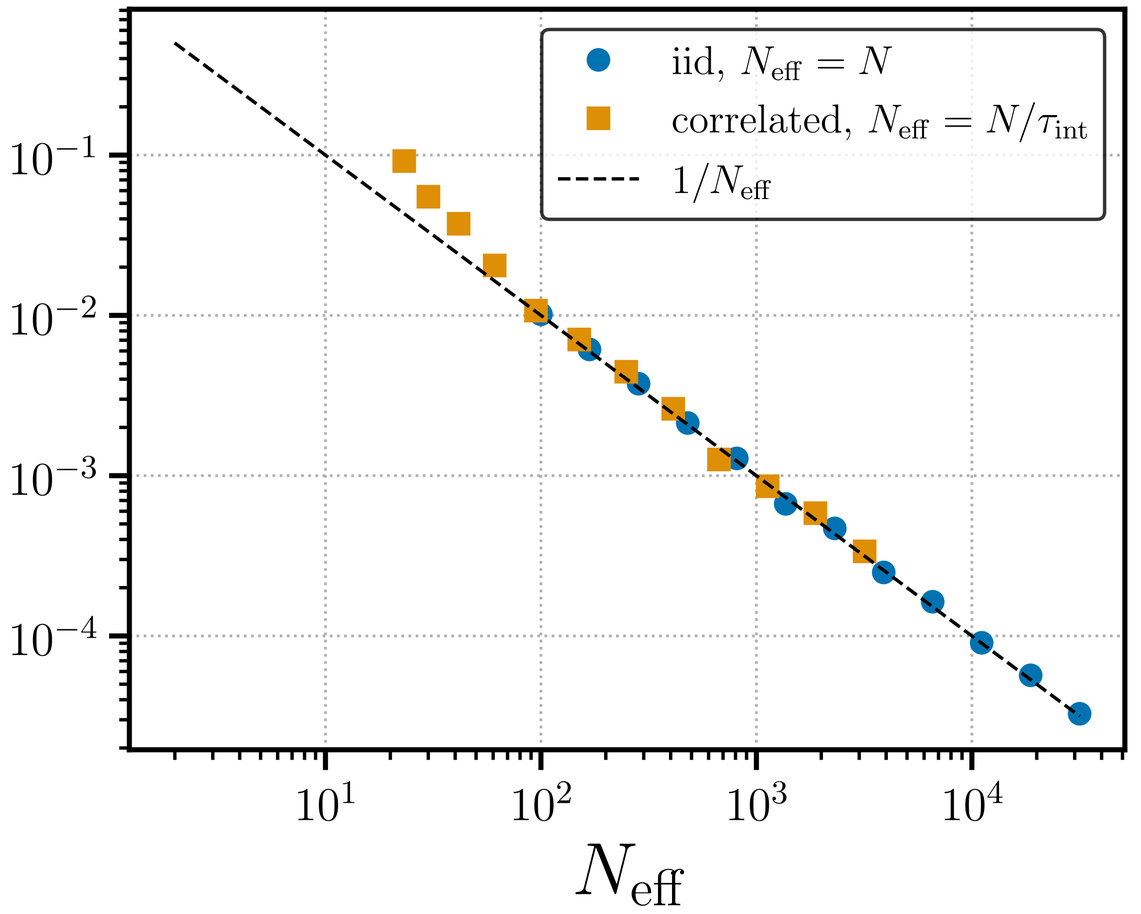


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Autocorrelation and effective sample size

Example





A reminder of our workflow

- Model the data generating process probabilistically.
- Condition the generative process on the observed data.
- Check that the results fit the observed data and improve the model.
  - Compare different models.

Model Checking

- First step: plot the model against the data.
  - Best fit model is ok
  - Model predictive (or posterior predictive) is better!
  - Sometimes obvious issues will stand out at this stage
- **Next step:** a χ^2 test

Chi-Square Goodness-of-fit

- If the data are Gaussian:

$$\vec{y} \sim \mathcal{N}\left(\vec{\mu}(\vec{\theta}), \Sigma(\vec{\theta})\right)$$

- Then a Chi-square test is a traditional choice to test how well the model fits the data.

Chi-Square Goodness-of-fit

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$$\vec{y} \sim \mathcal{N}\left(\vec{\mu}(\vec{\theta}), \Sigma(\vec{\theta})\right)$$

- Then a Chi-square test is a traditional choice to test how well the model fits the data.
- If we have some best fit parameters $\vec{\theta}^{\star}$, then the following statistic (which we call "the chi-square statistic"):

$$\chi^2(\vec{y}, \vec{\theta}^{\star}) = \left(\vec{y} - \vec{\mu}(\vec{\theta}^{\star})\right)^T \Sigma^{-1}(\vec{\theta}^{\star})\left(\vec{y} - \vec{\mu}(\vec{\theta}^{\star})\right)$$

- Is distributed according to a "chi-squared distribution with ν degrees of freedom" (yes, it's a bit confusing)

Chi-Square distribution

- The "chi-square distribution with ν d.o.f." has a PDF given by the following:

$$p_\nu(x) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2} .$$

- And it has an associated CDF, we'll call $\Phi_{\chi^2_\nu}$.

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- And it has an associated CDF, we'll call $\Phi_{\chi^2_\nu}$.
- With this, we can calculate the probability that our statistic $\chi^2(\vec{y}, \vec{\theta}^\star)$ is higher than we'd expect for typical Gaussian residuals under many repeated experiments.

$$p = 1 - \Phi_{\chi^2_\nu}\left(\chi^2(\vec{y}, \vec{\theta}^\star)\right)$$

→ some structure of the data is not accounted for

- This is the "probability to exceed" or often colloquially, the "p-value." If this is small, then we have a poor fit to our data.

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Chi-Square and d.o.f.

- The number of degrees of freedom is $\nu = n_{\text{data}} - n_{\text{params}}$, where n_{data} is the dimensionality of the data and n_{params} is the number of constrained parameters.
- By fitting the model to the data, we reduce the scatter in the residuals $\vec{y} - \vec{\mu}(\vec{\theta})$, so the number of degrees of freedom is reduced from n_{data} by n_{params}.
- Finding n_{params} in a Bayesian context can be challenging because some parameters might be constrained by the prior, instead of by the likelihood. In this case, they do not reduce the degrees of freedom.

Brief interlude showing χ^2 test on MAP parameters

- This appears in the exercise notebook for this week.

What about with posteriors?

- The beauty of posteriors is that we have quantified our uncertainty on the parameters
- We should wrap this into our χ^2 test.
- We use the **posterior predictive distribution** to do this.

What about with posteriors?

- We draw replicated data sets

$$\vec{y}^{rep} \sim p(\vec{y}^{rep} \mid \vec{y}) = \int p(\vec{y}^{rep} \mid \theta) p(\theta \mid \vec{y}) \, \mathrm{d}\vec{\theta}$$

- For some test statistic $T(\vec{y}, \theta)$ (in this case the χ^2 statistic we calculated before), we want to know the probability to exceed ("Bayesian p-value"):

$$p_B = \Pr\left[T(\vec{y}^{rep}, \vec{\theta}) \geq T(\vec{y}, \vec{\theta})\right]$$

$$\equiv \int\int \mathbb{I}_{T(\vec{y}^{rep}, \vec{\theta}) \geq T(\vec{y}, \vec{\theta})} p(\vec{y}^{rep} \mid \theta) p(\theta \mid \vec{y}) \, \mathrm{d}\vec{y}^{rep} \mathrm{d}\vec{\theta}$$

- In practice we do this Monte Carlo style. For S total iterations we keep track of N_{exceed}, starting from zero.
  - $\vec{\theta}^s \sim p(\theta \mid \vec{y})$ (drawn from our posterior)
  - $\vec{y}^{rep,s} \sim p(\vec{y}^{rep} \mid \vec{\theta}^s)$ (this is now a draw from the posterior predictive)
  - Check $T(\vec{y}^{rep,s}, \vec{\theta}^s) \geq T(\vec{y}, \vec{\theta}^s)$. If it's true, add 1 to N_{exceed}. If it's false, add 0.
- At the very end, we calculate: $p_B = N_{exceed}/S$.

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Some notes about p_B

- This now can be considered “extreme” if p_B is near 0 **or** near 1, because it means our replicated data is different from our real data in either direction.
- The choice of T is up to you!! If the model contains some specific, defining feature, you can define T to capture that feature.
- See **Chapter 6** of **Bayesian Data Analysis** for a *very very good* discussion of this.

Exercise 3 — Calculate p_B on the linear vs. quadratic models

Model Selection

Which model is better?

- Next week we'll talk about several different methods and paradigms for doing model selection